

88/50

ABERDARE TECHNICAL COLLEGE

UNIT OF COMPETENCY: DESIGN COMPUTERIZED

DATABASE

LEVEL: 5

56%

96%

UNIT CODE: 0612 451 08A

ELEMENT COVERED: DESIGN COMPUTERIZED DATABASE

CAT 1, TERM 3

TIME: 1 HOUR 30 MINUTES

TOTAL MARKS: 50

Instructions to Candidates

- ✓ Write your name and admission number in the space provided.
- ✓ This paper has **THREE** sections: A, B, and C. Answer **ALL** questions in all sections.
- ✓ Write your answers legibly in the spaces/answer booklet provided.
- ✓ Do not write on the question paper.

Name: KAIRLI MARCUS Admission no. 878 Date:

...14/09/2026...sign. [Signature].....

1 Database: A structured collection of related data that is stored and accessed electronically. ✓ 81

DBMS (Database Management System): A software that allows users to create, store, retrieve, update and manage data in database. ✓ 81

Data redundancy: The unnecessary repetition of the same data in database. ✓ 81

Data independence: → The ability to change structure of a database without affecting the application programs that use the data. ✓ 81

2 Data storage, retrieval and update. ✓

Data security and access control. ✓

Data integrity. ✓

Concurrent access and Multiuser support. ✓ 84

3. Banking: managing customer accounts, transactions & balances. ✓

Hospitals: → storing patient records and medical history. ✓ 84

Education: → Student information system and registration. ✓

Retail: → Product catalogues, customer orders and inventory. ✓

4. Centralized DBMS: All data is stored and managed from a single central location. ✓

Distributed DBMS: Data is stored across multiple locations but appear to user. ✓ 81

a) one single database. ✓

5 Advantages of DBMS over traditional file based system. ✓

Security: Database management system is much secure than the traditional file based system. ✓

Reduced data redundancy: data is stored once and stored avoiding unnecessary duplication. ✓ 84

6 Hardware: The physical devices that store and process the data. ✓

Software: The DBMS software itself plus operating system and application programs that interact with the database. ✓

Data: → The actual data stored in database. ✓ 81

User ✓ - People who interact with the system.

Process ✓ - The roles & instructions and includes controlling how database is used and managed. 09

7. Presentation tier ✓ - The front end layer where users interact with the system.

Application tier ✓ - Contains the business logic and application programs that process user request and communicate with database.

Data tier ✓ - The back end layer that stores and manages the actual data using the DBMS. 06

8. a) Elicitation requirements: 1) Process of gathering documents and understanding the data needs and functional requirements of users for database system.

b) Interviews with users & stakeholders.
Questionnaire. ✓ 02

9. Description of the data to be stored ✓

Organization of data and relationships ✓ 03

Constraints and security requirements.

	Relational database model	Hierarchical database model
Structure of data	Data is represented in form of tables ✓	Data is represented in a tree like structure ✓
Relationship of entities	Entities relationship are many to one and many to many ✓	Relationship is parent to child only ✓
Flexibility in querying data	Very flexible → Solving complex queries primary direction ✓	Limited flexibility → queries and plus predefined hierarchical path. 07

- Hierarchical can be used in banks for customer account management.
- Relational database model can be used in e-commerce website like for MySQL and others.

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